

Article

Exploring Long-Term Clean Energy Transition Pathways in Ghana Using an Open-Source Optimization Approach

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Abstract

Access to clean and sustainable energy technologies is critical for all nations, particularly developing countries in Africa. Ghana has committed to ambitious greenhouse gas emission reduction targets, aiming for 10% and 20% variable renewable energy integration by 2030 and 2070, respectively. This study explores potential pathways for Ghana to achieve its renewable energy production targets amidst a growing energy demand. An open-source energy modelling tool was used to assess four scenarios accounting for current policies and additional alternatives to pursue energy transition goals. The scenarios include Business as Usual (BAU), Government Target (GT), Renewable Energy (REW), and Net Zero (NZ). The results indicate that total power generation and installed capacity would increase across all scenarios, with natural gas accounting for approximately 60% of total generation under the BAU scenario in 2070. Total electricity generation is projected to grow between 10 and 20 times due to different electrification levels. Greenhouse gas emission reduction is achievable with nuclear energy being critical to support renewables. Alternative pathways based on clean energy production may provide cost savings of around USD 11–14 billion compared to a Business as Usual case. The findings underscore the necessity of robust policies and regulatory frameworks to support this transition, providing insights applicable to other developing countries with similar energy profiles. This study proposes a unique contextualized open-source modelling framework for a data-constrained, lower-middle-income country, offering a replicable approach for similar contexts in Sub-Saharan Africa. Its novelty also extended towards contributing to the knowledge of energy system modelling, with nuclear energy playing a crucial role in meeting future demand and achieving the country's objectives under the Paris Agreement.

Keywords: OSeMOSYS; clean energy; GHG emissions; RE integration; Ghana



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1. Introduction

The global push towards sustainable development and climate change mitigation has intensified the demand for effective energy planning, especially in developing nations,

such as Ghana. According to the Tracking SDG7 Report 2023 [1], the global electricity access rate was approximately 91% in 2021. Despite this progress, 675 million people worldwide, predominantly in Sub-Saharan Africa, still lack access to electricity. The global energy crisis has become significantly more complex, with a critical need to transition to net-zero emissions by 2050 while maintaining affordable and secure energy services [2]. As Ghana strives to meet its Nationally Determined Contributions (NDCs) under the Paris Agreement, it has encountered substantial challenges in transitioning from a dependence on fossil fuels to a sustainable, low-carbon energy system [3]. This transition is vital for reducing greenhouse gas (GHG) emissions, enhancing energy security, and fostering economic growth [4]. However, the complexity of energy systems and the socioeconomic context in developing countries require sophisticated modelling tools that can accommodate diverse scenarios and stakeholder interests [5].

Traditional energy models often lack the flexibility and transparency needed for inclusive decision-making processes [6]. In addition, some of the modelling software can be prohibitively expensive and inaccessible for local researchers and policymakers [7]. Therefore, there is an urgent need for open-source energy planning models that offers a robust, adaptable, and transparent approach to energy system modelling [8]. Open-source energy system optimization tools are increasingly used for analyzing long-term decarbonization pathways and assisting decision-makers and governments in policy formulation [5]. These models enable the integration of renewable energy sources, assessment of energy efficiency measures, and evaluation of various policy impacts [9]. By providing detailed insights into energy demand and supply dynamics, these tools support the design of effective strategies for sustainable energy transitions [10].

In developing countries, the adoption of open-source energy modelling tools can democratize access to advanced analytical capabilities, fostering local expertise and empowering stakeholders to participate in energy planning processes [11]. This is essential for addressing the unique challenges and opportunities within these regions, ensuring that energy policies are contextually relevant and socially equitable [12]. In Ghana, for instance, leveraging open-source optimization tools can facilitate the identification of optimal energy pathways, balancing economic, environmental, and social objectives [13]. By incorporating local data and stakeholder inputs, these models can enhance the accuracy and relevance of energy planning, ultimately contributing to the achievement of Ghana's NDCs and broader sustainable development goals [14].

Ghana, like many other developing countries, has committed to ambitious goals of achieving net-zero emissions by 2070, as outlined in its latest Energy Transition and Investment [3]. Despite having a higher electricity access rate compared to neighboring countries, Ghana has faced power interruptions, leading to load shedding, resulting in a deficit between available capacity and electricity demand [15]. The country's energy supply is predominantly reliant on traditional fossil fuel power generation technologies, which contribute significantly to annual greenhouse gas (GHG) emissions. The share of RE sources in power generation remains relatively low and less than 1%, highlighting the urgent need to diversify the energy supply mix and incorporate more clean energy resources to achieve a sustainable energy system [16].

This study intends to develop a comprehensive optimization model tailored to Ghana's unique energy landscape. By integrating local data, policy goals, and socioeconomic factors, the model aims to explore viable energy transition pathways and support informed decision-making for sustainable development. The Open-Source Energy Modelling SYStem (OSeMOSYS) is used to assess different alternative scenarios that include energy interventions, such as current policies, nationally determined contributions, sustainable growth priorities, and decarbonization efforts. Four scenarios are explored, including the Business

as Usual (BAU) scenario, the Government Target (GT) scenario, the 48% RE integration (REW) scenario, and the Net Zero (NZ) scenario. The study's findings are crucial in assisting policymakers and stakeholders in the energy sectors of Ghana to make informed decisions to accelerate the sustainable energy transition. The study makes significant contributions to advancing sustainable energy transitions in Ghana by providing actionable insights and strategies for policymakers and stakeholders. Through innovative modelling and analysis, the study answers the following research questions: (1) What are the optimal supply mix scenarios for Ghana's future power system that can ensure adequate, reliable, and sustainable electricity generation with high renewable energy penetration? (2) What role can nuclear power play in contributing to Ghana's future energy mix, particularly in terms of reliability, sustainability, and decarbonization? (3) What is the least-cost, optimal power supply mix that would enable Ghana to achieve net-zero emissions by 2070, considering detailed demand and supply dynamics?

Furthermore, in answering these questions, the study proposes a unique contextualized open-source modelling framework for a data-constrained, lower-middle-income country, offering a replicable approach for similar contexts in Sub-Saharan Africa. The novelty of the study is also extended towards contributing to the knowledge of energy system modelling, with nuclear energy playing a crucial role in meeting future demand and achieving the country's objectives under the Paris Agreement. Beyond Ghana, the modelling approach is highly transferable to other developing countries, particularly those with limited data availability and similar developmental challenges. As such, it offers a valuable framework to researchers for exploring evidence-based national and regional strategies to accelerate sustainable energy transitions and support long-term development planning in low-resource settings.

1.1. Current Energy Situation in Ghana

Ghana has one of Africa's highest rates of access to electricity. According to the Ghana Energy Statistics of 2023 [16], the national electricity access rate reached approximately 88.8% in 2022. Additionally, the share of the population with access to clean cooking technologies has significantly increased, rising from 15% in 2015 to 30% in 2022 [17]. Ghana's energy supply is sourced from a mix of natural gas (NG), oil, traditional biomass, hydro, and renewable energy (RE) [18]. As depicted in Figure 1, the share of each energy source in the total energy supply has varied over the years. Additionally, the total energy supply has doubled over the past two decades, growing from 257.36 PJ in 2000 to 516.23 PJ in 2022 [16]. This substantial increase underscores the growing energy demand and the need for a more diversified and sustainable energy mix to meet the country's future energy needs.

In 2022, Ghana's energy mix comprised 33.63% oil, 32.38% biomass, 28.15% natural gas, 5.7% hydro, and 0.11% solar photovoltaic (PV). The Energy Commission of Ghana (ECG), as the main energy authority, oversees and manages the development and utilization of energy resources to ensure affordable and secure energy supplies. In partnership with the Energy Commission, GRIDCo is responsible for the development of transmission and distribution networks across the country. Through this collaboration, electricity is generated and distributed, and Ghana also exports electricity to neighboring countries, like Benin and Togo. Ghana's final energy consumption includes electricity, biomass, and petroleum products, like gasoline, diesel, and LPG. As illustrated in Figure 2, final energy consumption has increased from 229 PJ in 2000 to 367 PJ in 2022. In 2022, petroleum products dominated final energy consumption, accounting for 49.2% of the total. The residential and transport sectors are the largest consumers, each accounting for 38% of total energy use, as shown in Figure 3. Industry consumed 19% of the energy, while the service sector accounted for 4%,

and agriculture for only 1%. The installed generation capacity is estimated to be 5454 MW in 2022, dominated by thermal power plants. Hydropower plants represent 29% of the installed capacity in 2022 (Figure 4).

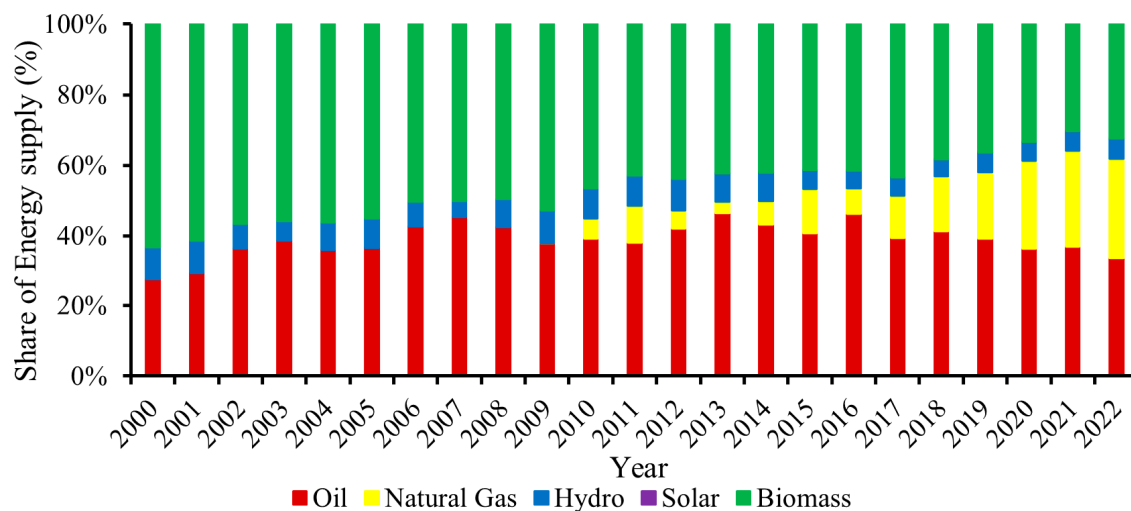


Figure 1. Trend in total energy supply from 2000 to 2022 [16].

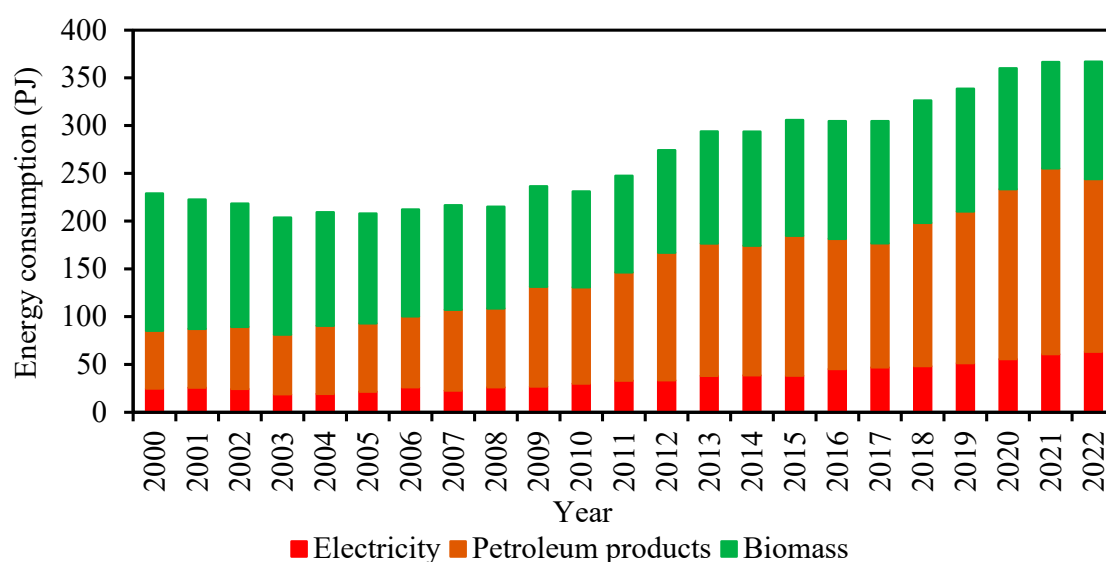


Figure 2. Trend in final energy consumption from 2000 to 2022 [16].

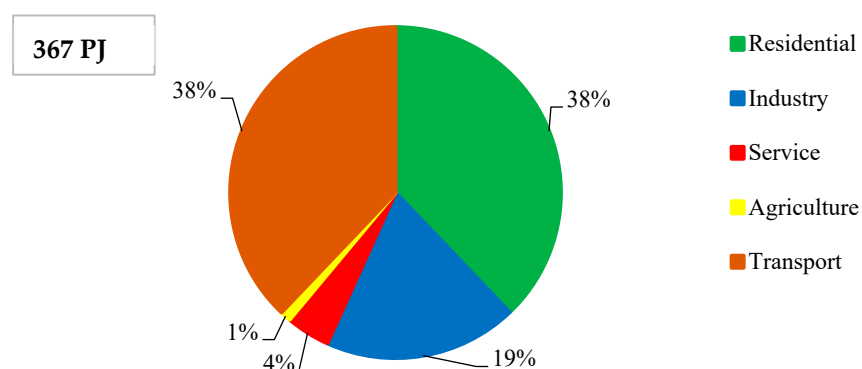


Figure 3. Final energy consumption per sector in 2022 [16].

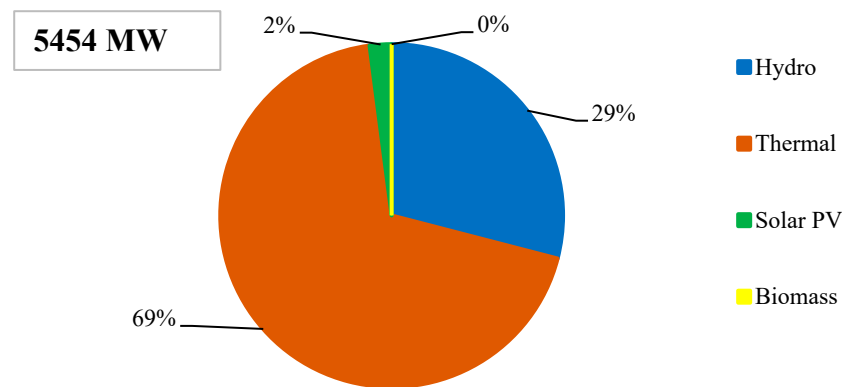


Figure 4. The installed power generation capacity in 2022 [16].

Figure 5 illustrates the trends in power generation in Ghana from 2000 to 2022. The electricity generation in Ghana has been primarily derived from hydro and thermal sources. The total electricity generation increased from 7859 GWh (28.3 PJ) in 2001 to 23,163 GWh (83.4 PJ) in 2022. Large hydropower was the leading source of electricity generation in Ghana between 2000 and 2015, with its share ranging from 51% to 92%. However, since 2016, the share of thermal power has increased steadily, reaching 64% in 2022. Ghana has extensive renewable energy (RE) policies to encourage renewable energy development. The main RE sources considered under these policies include solar, wind, biomass, tidal, landfill gas, geothermal, sewage gas, and small hydro (≤ 10 MW) [15]. The country has abundant RE potential that can be harnessed to boost its energy transition. Daily solar radiation ranges from 4 kWh/m² to 6 kWh/m² with an annual estimated solar potential of 28,343 PJ. Wind and biomass energy potential are estimated to amount to 219 and 119 PJ, respectively [19]. Despite this, the variable RE share in the supply mix is still low, less than 1%. It has, therefore, become crucial to develop strategies to help Ghana accelerate its transition to clean energy technologies.

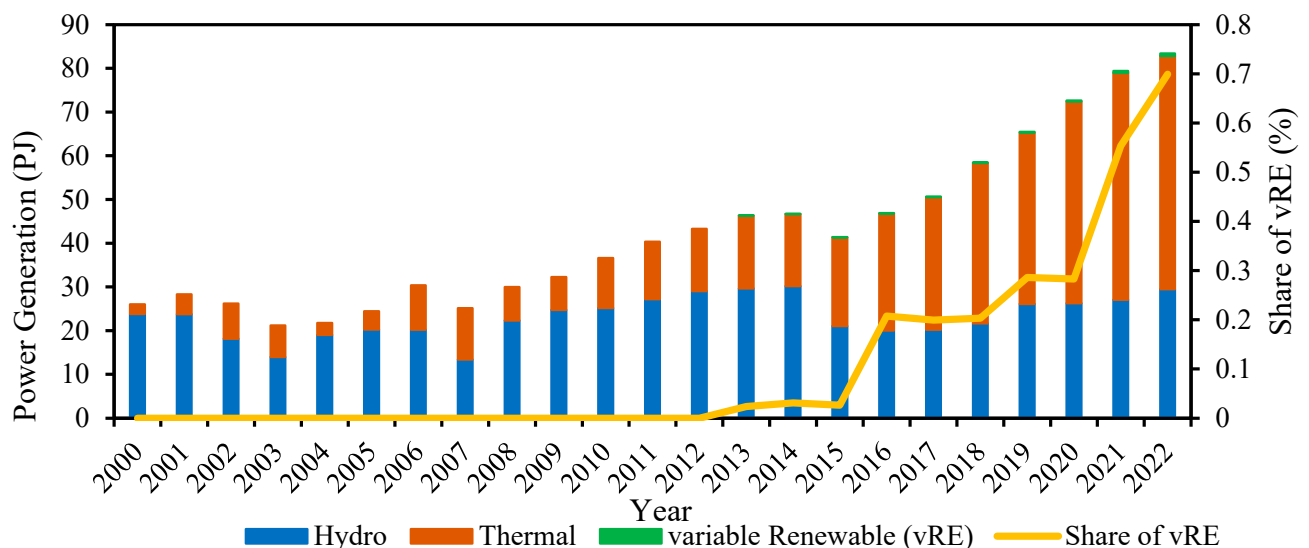


Figure 5. Trend in power generation from 2000 to 2022 [16].

1.2. Literature Review

Several authors have analyzed decarbonization pathways to inform clean energy policy in developing countries context. For instance, Plazas-Niño et al. [8] have used the OSeMOSYS model to assess plausible ways to decarbonize the energy sector of Colombia. The study's findings revealed that the carbon intensity of the energy sector could be

reduced by about 93% with considerable reduction in terms of energy intensity reduction, fuel imports, and socioeconomic benefits if adequate energy policies are developed. In addition, the same authors similarly demonstrated the usefulness of the OSeMOSYS tool in low-emission hydrogen roadmap planning in Colombia. They presented a techno-economic assessment of green hydrogen pathways to suggest full hydrogen economy deployment in the country [20]. In Ethiopia, Gabremeskel et al. [21], conducted long-term electricity supply modelling using OSeMOSYS. Their results showed that RE technologies are more competitive and favorable in the Ethiopian context because of their low cost and abundant availability. Hydropower is found to play a crucial role in the future energy system of Ethiopia in addition to solar PV, CSP, wind energy, geothermal, and natural gas.

Giuha Manco et al. [22] reviewed various approaches to multi-energy system modeling and optimization, discussing six distinct cases and presenting a comprehensive mathematical framework as a reference for developing energy models. By synthesizing various modeling techniques, the authors provided valuable methodological insights for building robust and scalable energy modeling approaches. Similarly, Bidattul Syirat Zainal et al. [23] explored advancements in green hydrogen production technologies. Their review highlighted the cost-effectiveness of producing green hydrogen using renewable energy sources, such as solar and onshore wind. The integration of these renewables with electrolysis was shown to be both beneficial and cost-efficient [24]. The authors also proposed strategies for scaling up green hydrogen production, emphasizing its potential as a sustainable energy carrier and its critical role in reducing greenhouse gas (GHG) emissions. Furthermore, a long-term energy system modeling approach was employed to evaluate clean energy transition pathways in Egypt [25]. Using the OSeMOSYS tool, the authors analyzed multiple scenarios and identified solar PV and onshore wind as key technologies for scaling up to achieve the country's clean energy transition objectives.

Ibrahim et al. [26] reviewed the possibility of energy supply through RE production in selected African countries. The results revealed the availability of huge RE potential in African countries, such as Nigeria, Cameroon, Ghana, and South Africa. Furthermore, it is found that, aside from South Africa, the other countries have failed to exploit the RE resources, which are crucial for transitioning to a more sustainable energy system. Akpahou et al. [27] developed strategies for sustainable energy planning in the Benin Republic. By combining natural gas with solar PV, CSP, wind, and hydropower, the authors have developed different scenarios crucial for the country to integrate more renewables into the national grid to achieve their NDC under the Paris Agreement. Similarly, using the OSeMOSYS framework, Pusania et al. [28] investigated pathways to clean energy transition in Indonesia. Their findings demonstrated that coal plant retirement and large-scale RE deployment constitute crucial pathways to achieving a clean energy transition in the country.

Nyasapoh et al. [29] evaluated the effectiveness of clean energy technology adoption in Ghana. By employing the MESSAGE model, the authors have demonstrated that it is crucial to transition from fossil fuel-based technologies to clean energy resources. Despite an increase in capital investment cost, mitigation measures are found to reduce emissions by 15%. Similarly, Gadzanky [30] developed a comprehensive framework to assess the electricity expansion plan in Ghana. The study results revealed several challenges related to power system reliability and resilience in the crude oil and natural gas supply. Strategies, such as diversification of the supply mix and flexible electricity generation, are recommended for a stable power supply. Afful-Dadzie et al. [31] developed a generation expansion planning model to assess renewable electricity generation targets in developing countries context. The authors have used the model in the case of Ghana and found that if the country intends to meet its target of 10% of variable RE by 2030, it needs to invest

more than 1% of its GDP. The study further highlights the importance of the integration of RE technologies and the policy needed to boost the clean energy transition in Ghana. In addition, Odonkor et al. [32] assessed public perception and acceptance of nuclear power deployment in Ghana. The authors found that about 51% of Ghanaians responding to their survey support Ghana's nuclear power plant development plans.

Samuel Gyamfi et al. [33] investigated the potential of renewable energy in improving electricity supply security in Ghana. RE sources, such as mini-hydro, solar PV, wind, and bioenergy, are abundantly available in the country and, if well managed, can substantially increase the share of the electricity supply mix. The authors highlight the importance of investigating the possibility of supplying electricity from RE generation technologies to meet demand and diversify the supply mix. Similarly, Nyarko et al. [34] assessed the implication of RE penetration in the electricity supply mix of Ghana by considering three levels (10%, 20%, and 30%). While the authors highlighted the benefits of integrating renewable energy into the supply mix, the study recommends exploring the techno-economic feasibility or the policy measures required to implement the proposed scenarios. In addition, Vazdanie [35] used an optimization model to conduct resilient energy system analysis in Ghana and China. The author conducted various analyses to demonstrate the integration of resilient energy systems in developing regions and their associated benefits. However, the study acknowledges several limitations and recommends further analyses to enhance the results. Merem et al. [36] assessed the use of RE technologies to meet the growing electricity demand in the power sector of Ghana. The study highlights several challenges hindering the large-scale deployment of renewable energy in Ghana, along with their socioeconomic and environmental impacts. However, it falls short of proposing comprehensive scenarios and conducting the demand-supply analysis necessary for achieving a sustainable electricity transition in the country.

John Bosco et al. [37] modelled energy consumption in Ghana's informal sector to evaluate its impact on national energy demand. The study employed the autoregressive distributed lag estimation technique to quantify the relationship between the informal sector and overall energy consumption. Additionally, suppressed electricity demand was estimated considering climate change conditions, the informal economy, and sector inefficiencies [38]. Using a quantile autoregressive distributed lag approach, the study forecasted suppressed electricity demand, providing valuable insights for energy system planning in developing countries. Similarly, Ebenezer K. et al. [39] assessed flood risk under shared socioeconomic pathways in Ghana. The authors analyzed flood susceptibility and its impact on electricity bulk supply points using a frequency ratio model, incorporating various flood conditioning factors. The findings highlighted the vulnerability of electricity infrastructure, indicating that the risk of electricity disruptions due to flooding is expected to increase in the future. Juan Carlos et al. [40] utilized the LUT Energy System Transition Model to design a 100% renewable-based energy system for Chile. Their findings indicate that by 2050, renewable electricity generation will be primarily driven by solar PV and wind energy, with installed capacities reaching 43.6 GW and 24.8 GW, respectively, alongside reduced energy costs. In addition, Takashi Otsuki et al. [41] applied the NE Japan Energy System Optimization Model to assess the impact of carbon capture and storage (CCS) uncertainties on Japan's pathway to net-zero emissions. The model effectively quantified the role of CCS in emission reductions and its influence on the integration of net-zero carbon fuels, such as hydrogen.

Moreover, Saad et al. [42] have employed the OSeMOSYS tool for improved energy security and long-term energy system modelling in Botswana. The findings underscored the need for significant financial investment to boost RE deployment. Adequate RE policies have been found crucial, including the adoption of strategies for solar PV and storage

expansion, and updated regulatory frameworks, to foster decarbonization and ensure universal access to energy. Similarly, Awopone et al. [43] have used OSeMOSYS to assess optimal pathways for generation systems in Ghana. Their results suggested that significant economic and environmental benefits could be achieved by employing adequate energy efficiency measures. However, the study did not investigate the potential integration of nuclear plants into the future energy supply mix or analyze the long-term evolution of end-use sector electrification. An assessment of solar energy policy implementation in Ghana has revealed that appropriate RE policy could drive solar technologies development and reduce the dependence on conventional energy resources [15]. Jaime et al. [44] employed the Urban Building Energy Model (UBEM) to design zero-carbon energy communities in low-industrial areas on Corvo Island. Their results were validated and enhanced through comparison with simulations using the EnergyPLAN model, incorporating local renewable energy sources. The study highlights the significance of urban energy analysis and the benefits of model coupling for optimizing sustainable energy systems. Similarly, Damiele et al. [45] developed an economic optimization approach for sizing renewable energy mixes on small islands. By using a tailored GIS-based data collection tool, their results demonstrated a lower levelized cost of energy (LCOE) and a higher share of renewable energy generation.

The studies reviewed above have demonstrated that OSeMOSYS and energy modelling are important in assisting policy formulation and decarbonization pathways analysis in developing countries. However, there are several limitations and gaps in the reviewed literature regarding the modelling approach and electricity transition analysis in the energy sector of Ghana. These include a lack of detailed techno-economic feasibility analyses, insufficient exploration of necessary policy frameworks, and limited study on the demand-supply scenario analysis with an open-source optimization model. This study applies an open-source optimization approach to model Ghana's long-term energy system, incorporating nuclear energy as a potential clean resource in the future energy mix. The study attempts to assist researchers, policymakers, and authorities in the energy sector's attention to the potential benefits that could be attained by transitioning from traditional energy sources to clean energy technologies to call for further investment in the sector. It fills the identified gaps by providing a detailed techno-economic feasibility analysis with up-to-date country-specific data, and contributes to the body of knowledge on scenario analysis with an open-source optimization model. Delivers actionable policy pathways tied to the Data-to-Deal framework, with robust validation, stakeholder engagement, and context-specific techno-economic insights. To meet the growing energy demand and promote sustainability, Ghana aims to diversify its energy sources by increasing the share of renewables, such as solar, wind, bioenergy, and small hydro.

The government is actively working on policies and incentives to attract investment in renewable energy projects. Additionally, efforts are being made to improve energy efficiency across all sectors, which is crucial for reducing greenhouse gas emissions and achieving the country's climate goals, demonstrating the need to conduct the present analysis. Nuclear can provide cost-effective baseload low-carbon power, with a targeted installed capacity of at least 3 GW starting from 2045, as stated in the Ghana Energy Transition and Investment plan [3].

In addition, OSeMOSYS is an ideal tool for energy modeling in developing countries, like Ghana, due to its accessibility, flexibility, and cost-effectiveness. Unlike proprietary tools, such as TIMES or MESSAGE, OSeMOSYS is free, open-source, and easy to use, with minimal computational and data requirements, making it suitable for resource-constrained environments [46]. It adheres to the U4RIA principles, emphasizing transparency, reproducibility, and interoperability, which are crucial for trust and collaboration in energy

modeling. With its growing global community, extensive documentation, and ability to scale from simple to complex systems, OSeMOSYS provides a practical and robust framework for long-term energy planning in developing contexts [5].

Hence, the remaining sections of the study are outlined as follows: Section 2 presents the methods and materials used to achieve the objectives of the study including a brief description of the OSeMOSYS framework. Section 3 presents the results and discussions, while Section 4 provides recommendations and concluding remarks.

2. Materials and Methods

This section describes the methodological approach used in analyzing clean energy transition decarbonization pathways in Ghana. The Ghana Starter Data Kit [19] and Ghana base SAND file in the Supplementary Materials [47] are used as primary data sources and references for further developing the OSeMOSYS-Gh model and parameterizing the model.

2.1. Model Development and Reference Energy System

This subsection provides an overview of the OSeMOSYS model and outlines the structure of the reference energy system for Ghana as represented in this study

2.1.1. OSeMOSYS Energy Modelling Tool

The OSeMOSYS software and its clicSAND version 3.0 have been employed in developing OSeMOSYS-Gh. This offers the unique advantage of developing different scenarios crucial for decision-makers and policymakers in the energy sector at no cost, in an open-source environment [9]. OSeMOSYS is a bottom-up optimization energy modeling system that uses a detailed representation of technologies and flows in the energy system, considering performance, costs, resource use, and environmental impacts [48]. OSeMOSYS, like other long-term optimization models, operates under the assumption of perfect foresight and perfect competition in energy markets [49]. This means that it assumes a clear understanding of future conditions and that energy markets function in a perfectly competitive manner. Mathematically, OSeMOSYS is a deterministic and linear optimization framework [21]. It has been widely utilized as a long-term optimization model in various countries [50]. It calculates the cheapest way of producing energy to meet a pre-defined demand given a set of power generation technologies. In its interface, technologies are defined by their costs, and technical parameters, such as capacity factor, lifetime, and production potential. OSeMOSYS core model objective function is given by Equations (1) and (2), which allow the minimum net present value cost to be determined over the lifetime period of a given technology. Different technologies compete to gain shares in the electricity supply based on their specific characteristic and imposed constraints [9].

$$\text{Minimize} \sum_{y,t,r} \text{TotalDiscountedCost}_{y,t,r} \quad (1)$$

$$\begin{aligned} \forall_{y,t,r} : \text{TotalDiscountedCost}_{y,t,r} = & \text{DiscountedOperatingCost}_{y,t,r} \\ & + \text{DiscountedCapitalInvestment}_{y,t,r} \\ & + \text{DiscountedTotalEmissionPenalty}_{y,t,r} \\ & - \text{DiscountedSalvageValue}_{y,t,r} \end{aligned} \quad (2)$$

where y , t , and r represent the year in time horizon, type of power plant technology, and region or country modelled, respectively. More information can be obtained on the detailed calculations behind the OSeMOSYS model in the OSeMOSYS documentation [51].

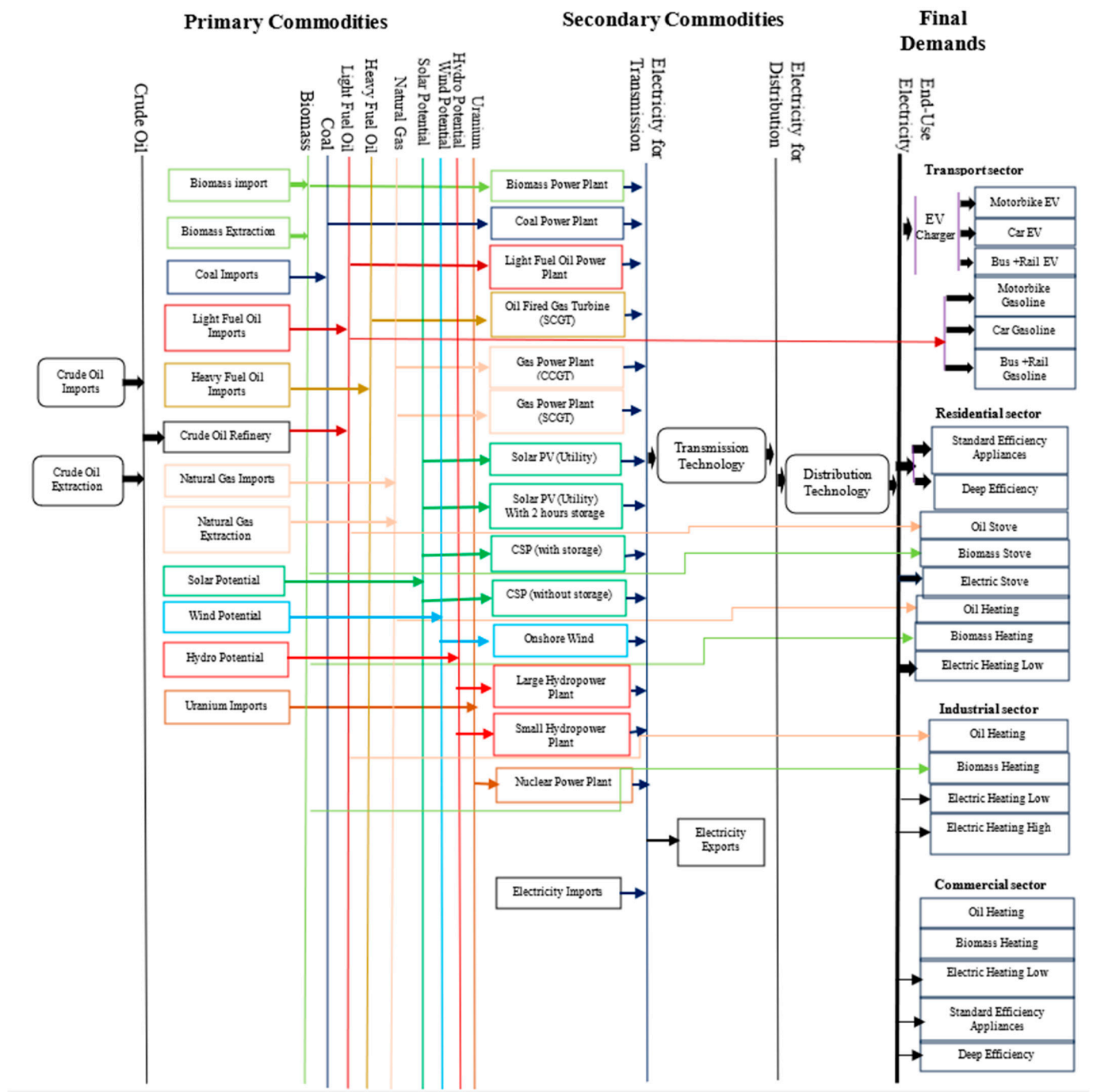


Figure 6. Reference energy system of Ghana.

Table 2. Electricity demand projection data (PJ).

Sector\Year	2022	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070
Industry	26.74	31.97	40.28	48.54	58.49	70.48	84.93	102.34	123.31	148.59	179.05
Residential	25.6	27.92	30.39	32.89	35.35	37.74	40.01	42.11	44.04	45.81	47.41
Commercial	10.67	11.89	14.98	18.06	21.76	26.22	31.59	38.07	45.87	55.28	66.61

Furthermore, each technology has its techno-economic parameters, such as capacity factor, fixed, variable, and capital cost, as well as operational lifetimes. Tables 3 and 4 illustrate the techno-economic data input into the OSeMOSYS-Gh model. The values presented in Table 3 include the projected capital costs and fixed operation and maintenance (O&M)

costs for various power generation technologies for key years: 2020, 2030, 2050, and 2070. These cost trajectories were directly integrated into the OSeMOSYS model as technology-specific investment and fixed cost parameters, influencing the least-cost optimization and the selection of technologies over time based on their economic competitiveness. In addition to cost data, Table 4 includes key technical parameters, such as efficiency, operational life, variable O&M costs, and capacity factors. Efficiency impacts fuel consumption and emissions; capacity factors determine actual energy output relative to installed capacity; operational life affects technology replacement timing; and variable costs contribute to overall system operating expenses. These inputs define the performance and utilization of each technology within the model. Fuel projection price and the methodology applied in obtaining data related to fuel and parameters can be found on Zenodo [47] and the Ghana Starter Data Kit [56].

Table 3. Capital and fixed costs for electricity generation technologies [47,57–61].

Technology	Capital Cost (USD/kW)				Fixed Cost (USD/kW)			
	2020	2030	2050	2070	2020	2030	2050	2070
Biomass Power Plant	5390.6	4489.13	3919.66	3878.75	157.22	157.22	157.22	157.22
Coal Power Plant	3549.41	3320.91	2749.75	2749.75	77.4	73.7	63	63
Light Fuel Oil Power Plant	1200	1200	1200	1200	35	35	35	35
Oil-fired Gas Turbine (SCGT)	1450	1450	1450	1450	45	45	45	45
Gas Power Plant (CCGT)	1247.53	1181.27	1015.55	1015.55	31.1	29.7	26.1	26.1
Gas Power Plant (SCGT)	1120.4	1049.52	88.1.23	872.4	24	22.9	20.3	20.3
Solar PV (Utility)	1290.8	1205.95	829.2	829.2	22.53	19.93	15.79	15.79
CSP Without Storage	4058	2634	2562	2562	40.58	26.34	25.62	25.62
CSP With Storage	5797	3763	3660	3660	57.97	37.63	36.6	36.6
Large Hydropower Plant	3000	3000	3000	3000	90	90	90	90
Small Hydropower Plant	3000	3000	3000	3000	90	90	90	90
Onshore Wind	1961	1790.18	1558.97	1558.97	30.3	28.65	26.58	26.58
Offshore Wind	3972.4	2450	2100	2100	158.89	98	84	84
Nuclear Power Plant	10,134.58	9459.2	9459.2	9459.2	118.8	118.8	118.8	118.8
Solar PV (Distributed with Storage)	2101.7	1969.3	1504.36	1504.36	70.7	65.65	60.15	60.15

Table 4. Techno-economic parameters for electricity generation technologies [37].

Technology	Operational Life (Years)	Efficiency	Average Capacity Factor	Variable Cost (USD/kW)
Biomass Power Plant	30	0.35	0.5	5.04
Coal Power Plant	35	0.37	0.85	0.0001
Light Fuel Oil Power Plant	25	0.35	0.8	0.0001
Oil-fired Gas Turbine (SCGT)	25	0.35	0.8	0.0001
Gas Power Plant (CCGT)	30	0.48	0.85	0.0001
Gas Power Plant (SCGT)	25	0.3	0.85	0.0001
Solar PV (Utility)	24	1	0.155	0.0001
CSP Without Storage	30	1	0.226	0.0001
CSP With Storage	30	1	0.264	0.0001
Large Hydropower Plant	50	1	0.54	0.0001
Small Hydropower Plant	50	1	0.54	0.0001
Onshore Wind	25	1	0.1	0.0001
Offshore Wind	25	1	0.1	0.0001
Nuclear Power Plant	50	0.33	0.85	3.13
Solar PV (Distributed with Storage)	24	1	0.181	0.0001

2.2.2. Model Assumptions

The primary model assumptions in this study are derived from the default settings and the Ghana base SAND model. The Ghana Starter Data Kit provides comprehensive energy data inputs and model assumptions for developing the OSeMOSYS energy system model for the country. These adapting assumptions, which serve as input parameters for the model's simulation, ensure that the scenarios accurately reflect real-world conditions in Ghana. The base data kit is frequently calibrated to align with current policies and local data sources, ensuring the model's assumptions are based on the most accurate and up-to-date information available. This calibration process is crucial for the robustness of the simulation, generating meaningful insights and reliable projections for Ghana's energy transition. Additionally, variable renewable energy sources are constrained within the model to ensure that the system can operate with high renewable shares while meeting the maximum share of total demand. In this study, four activity sectors have been modelled: the residential sector, the industry sector, the service sector, and the transport sector.

In the residential sector, for example, biomass is assumed to be used for cooking (70%) and low heating (30%). Oil is exclusively used for cooking. In the commercial and industrial sectors, biomass and oil are used for low heating only, whereas there is no electricity used for cooking or low heating. In all sectors, electrical appliances have a standard assumed efficiency equal to 1 for this model. Equations (3) and (4) are used to calculate final energy demand (FD) and technology residual capacity (TRC), as follows [61]:

$$FD = \sum_{\text{Techs}} \text{Energy Consumption (PJ)} \div \text{Input Activity Ratio} \quad (3)$$

$$TRC \left(\frac{\text{PJ}}{\text{year}} \right) = \text{Energy consumed by technology} \left(\frac{\text{PJ}}{\text{year}} \right) \div \text{Input Activity Ratio} \quad (4)$$

Regarding the progressive reduction in residual capacity, a constant existing capacity is assumed until 2025. Subsequently, we implement a linear decrease until reaching 0 PJ/year in 2035, which aligns with the average lifetime of residential technologies, typically ranging between 10 and 15 years for most technologies.

In the transport sector, three energy services are modeled: car transport demand, motorcycle transport demand, and bus transport demand. By utilizing historical data of fleet stock and the techno-economic parameters for each technology, we calculated the fuel energy consumption per technology (FECT) using Equation (5) [61].

$$FECT(PJ) = \frac{\text{Fleet stock } (v) \times A_f \times \text{Load factor} \left(\frac{P}{v} \right) \times \text{Energy intensity} \left(\frac{PJ}{Gpkm} \right)}{10^9} \quad (5)$$

where A_f = activity factor(km), E_i = energy intensity, $\frac{PJ}{Gpkm}$ = petajoules per giga person–kilometer.

The final transport demand and the TRC in the transport sector are then estimated using a similar approach in Equations (3) and (4). The final data can be found on the Zenodo repository [47].

2.3. Emission Factors

Fossil fuel technologies emit several greenhouse gases, including carbon dioxide, methane, and nitrous oxides throughout their operational lifetime. These are accounted for using GHG emission factors assigned to each fuel, rather than each power generation technology. The emission factor data were obtained from [62], and the equivalent emission factors in terms of CO₂e were calculated using the global warming potentials (GWP) outlined in [63]. For the modelling approach, emissions were calculated as a product of the

technology activity level and the emission factor, thus the emission factor by technology will depend on the efficiency of the technology [61]. Equations (5) and (6) are used to calculate the equivalent emission factor (Emf) and emission factor (Emt) per technology, respectively [61].

$$\text{Emf} \left(\frac{\text{MtCO}_2\text{e}}{\text{PJ}} \right) = \text{CO}_2 \text{ emission factor} \left(\frac{\text{Mt}}{\text{PJ}} \right) \times \text{CO}_2 \text{ GWP} \\ + \text{CH}_4 \text{ emission factor} \left(\frac{\text{Mt}}{\text{PJ}} \right) \times \text{CH}_4 \text{ GWP} \\ + \text{N}_2\text{O emission factor} \left(\frac{\text{Mt}}{\text{PJ}} \right) \times \text{N}_2\text{O GWP} \quad (6)$$

$$\text{Emt} \left(\frac{\text{MtCO}_2\text{e}}{\text{PJ}} \right) = \frac{\text{Fuel Emission factor Equivalent emission factor} \left(\frac{\text{MtCO}_2\text{e}}{\text{PJ}} \right)}{\text{Technology efficiency}} \quad (7)$$

2.4. Scenarios

As illustrated in Figure 7, four scenarios have been investigated: Business as Usual (BAU), the Government Target (GT) scenario, the 48% RE scenario (REW), and the Net-Zero scenario (NZ).

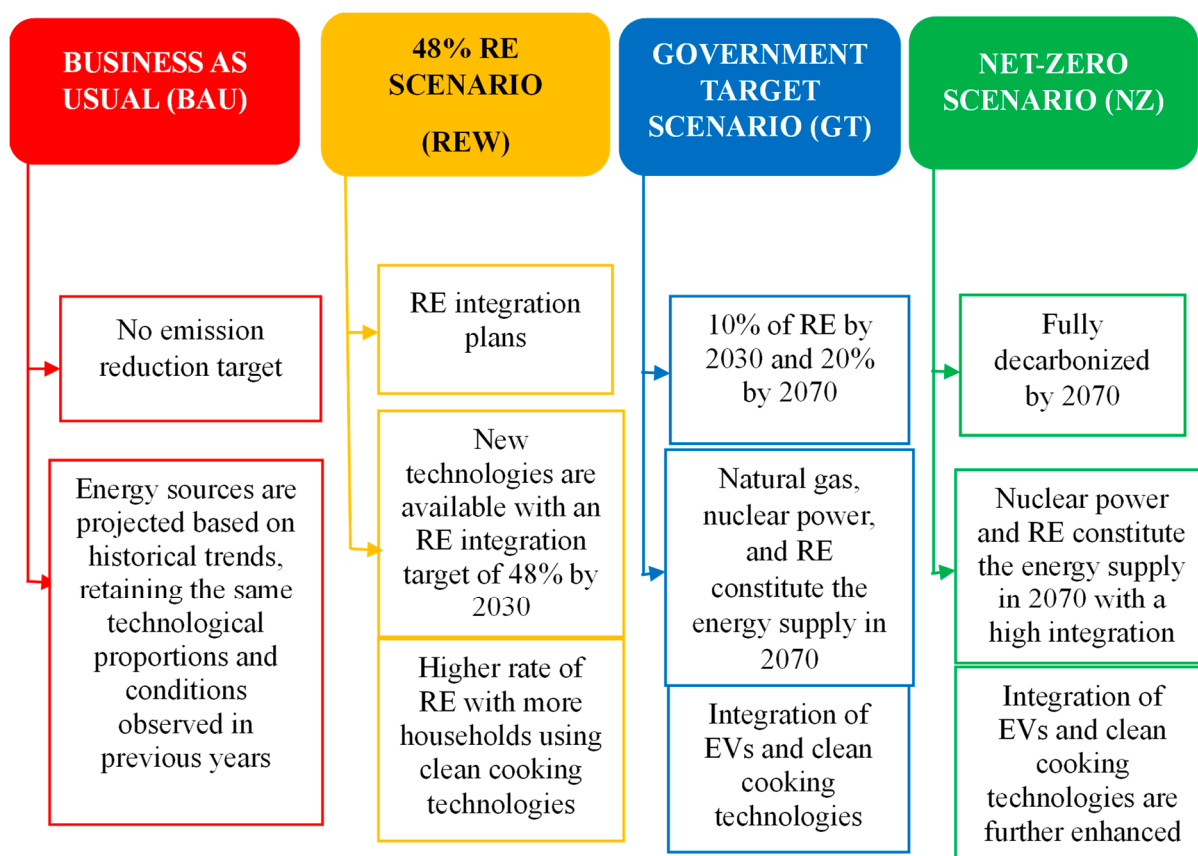


Figure 7. The four scenarios explored.

The Business as Usual (BAU) scenario is the scenario where there is no intervention from governmental policies. In this scenario, it is assumed that the primary energy consumption in Ghana will continue to be predominantly fueled by natural gas and biomass, while the existing challenges within the energy sector persist. Emission limits are not enforced, and energy sources are projected based on historical trends, retaining the same technological proportions and conditions observed in previous years. This scenario is useful as a comparative benchmark against alternative pathways.

The Government Target (GT) scenario explores the options of the country to meet the growing energy demand by 2070 while reducing emissions. It also considers the integration of 10% of variable renewable energy technologies by 2030, and 20% by 2070 in line with the goal stated in the 2021 NDC report of Ghana [4]. Nuclear power plant integration is set to start from 2030 onwards.

The 48% RE integration scenario is derived from the ECOWAS ambitious goal of achieving a robust 48% RE penetration by 2030. This scenario investigates the supply mix necessary for Ghana to achieve this goal at country level and the role of nuclear energy in the future of the energy matrix of Ghana [57].

The NET-ZERO scenario explores the energy mix while establishing a zero (0) emissions target by 2070, this being in line with the current stated goal in the [64]. Renewable energy represents at least 20% of the energy mix in this scenario. All technologies are available, including concentrated solar power plants (CSP), nuclear power plants, and wind and solar power plants. It should be noted that an additional restriction was applied, linearly lowering the activity of the CO₂-emitting power plants from 2030 onwards.

It is assumed that services in other sectors are electrified for all scenarios except the BAU, which follows the current trends and pace in the energy sector. It assumes that the country would shift from traditional biomass-based cooking stoves and gasoline-based vehicles to more environmentally friendly technologies, such as clean cooking stoves and electric cars.

2.5. Sensitivity Analysis

In OSeMOSYS-Gh, a discount rate of 10% is used as a baseline. To understand the sensitivity of the model to changes in the discount rate, a sensitivity analysis was conducted using four different rates: 5%, 8%, 12%, and 15%. This analysis aims to demonstrate the impact of variations in key input data and assumptions on power generation technologies.

3. Results

This section presents and discusses the main findings of the study. The outcomes of the four scenarios (BAU, GT, REW, and NZ) have been compared and analyzed to suggest clean energy transition pathways for Ghana.

3.1. Technology Deployment Analysis

Deploying new energy technologies with high RE integration is critical to diversifying the supply mix while fostering the transition to a clean and sustainable energy system. As displayed in Figure 8, the installed capacity for power generation has substantially increased in all scenarios between 2020 and 2070. If the total generation capacity is about 18.91 GW in 2070 under the BAU scenario, it reaches 96.7, 57.3, and 167.2 GW in the same year under the REW, GT, and NZ scenarios, respectively. This increase is mainly due to the integration of the RE technologies. As the capacity factor of intermittent RE technologies, such as CSP, solar PV, and onshore wind, is lower than conventional plants' capacity factors, it is necessary to install more capacities of these technologies to meet the same energy demand as in the BAU scenario. Furthermore, GT, REW, and NZ scenarios have the advantage of powering all energy demand sectors and end-uses, increasing the electrification rate. In the REW and GT scenarios, natural gas (NG) resources are kept in the mix with RE resources and nuclear energy due to the low emission factor, and Ghana considers NG a transition fuel. These results are similar to what is presented in the Ghana Energy Transition Framework, where installed capacity is projected to reach around 22 GW by 2070 in the No Policy Intervention (NPI) scenario [64].

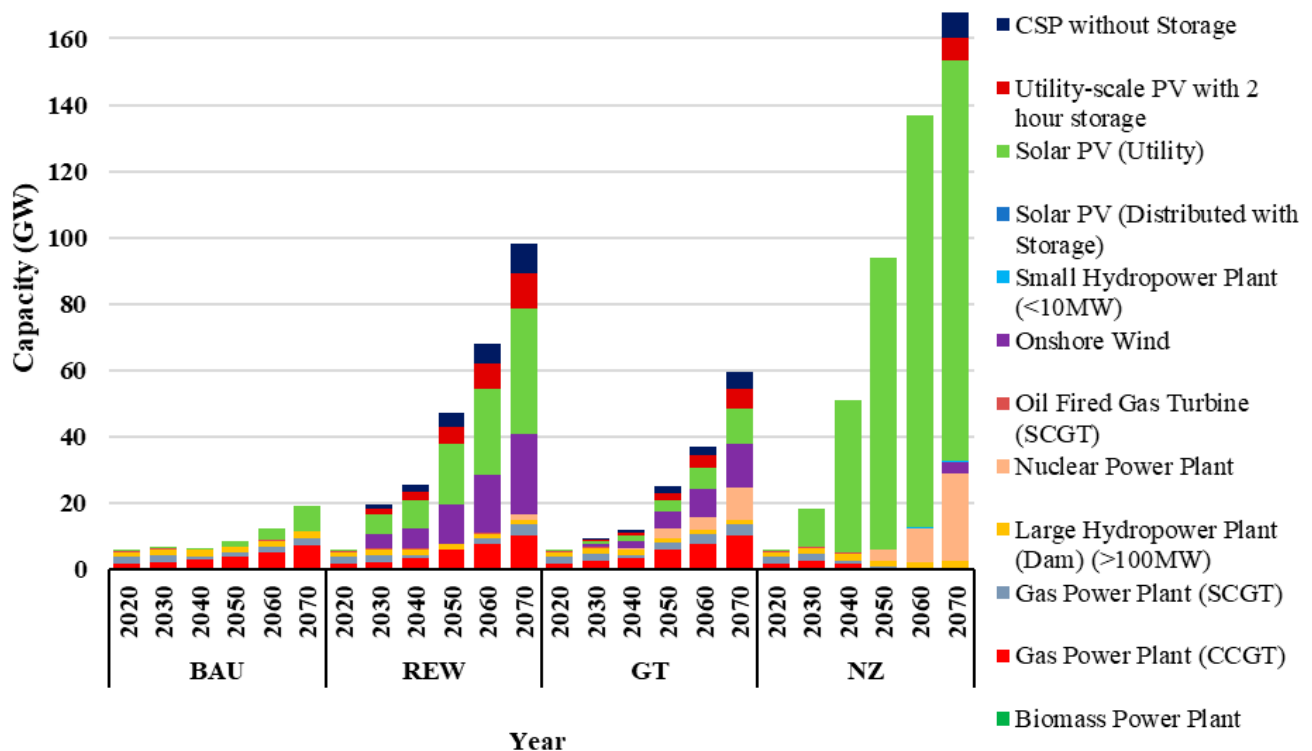


Figure 8. Installed electricity generation capacity (GW).

Figure 9 displays optimal capacity expansion plans of power generation (in PJ) from different energy technologies between 2020 and 2070 under the four scenarios. All scenarios demonstrate that, as energy demand increases, the power generation also increases. The total power generation under the BAU scenario increases to about 323.22 PJ, which is 4.3 times higher than the 74.71 PJ generated in 2020. The alternative scenarios' power generation is substantially higher, with a total generation of 770 PJ in 2070 in each of the REW and GT scenarios and 1500 PJ in the NZ scenario for the same year. The REW, GT, and NZ scenarios are expected to produce more power, increasing the electrification rate to meet all sectors' energy demands. With the current pace and trend in the energy sector of Ghana, natural gas sources would be the most dominant energy source constituting almost 60% of the power generation in 2070 in the absence of policy formulation (BAU scenario). The transition towards clean energy sources is done by the consideration of solar photovoltaic (PV), onshore wind, concentrated solar power (CSP), small hydropower, and nuclear power under all three alternative scenarios. The share of RE in the REW scenario has been high throughout the years with the consideration of nuclear and natural gas power plants. The power generation from the natural gas power plant represents about 35% in each of the GT and REW scenarios. In addition, nuclear power plant integration becomes visible by 2050 in the GT scenario and by 2060 in the REW scenario. These two policy scenarios have shown a more diversified energy supply mix, indicating a shift towards more resilient and sustainable energy compared to the BAU scenario, which relies heavily on gas and conventional energy sources. The NZ scenario, on the other hand, does not include natural gas or any CO₂-emitting technology. Nuclear power plants would generate 694.17 PJ by 2070, while intermittent RE technologies would produce 806.8 PJ of electricity by 2070. Utility-scale solar PV and onshore wind technologies are the most dominant RE technologies under this scenario.

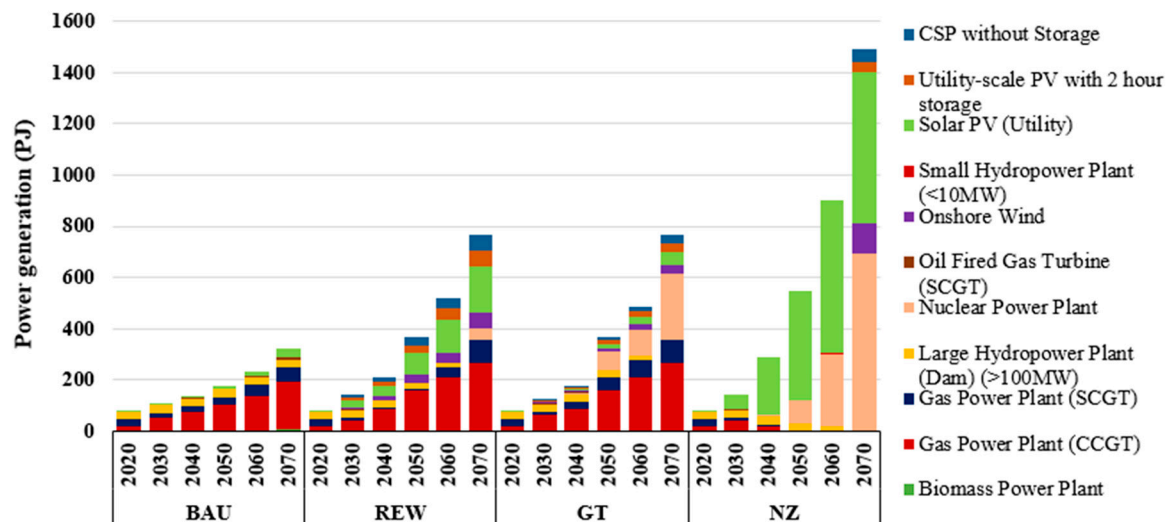
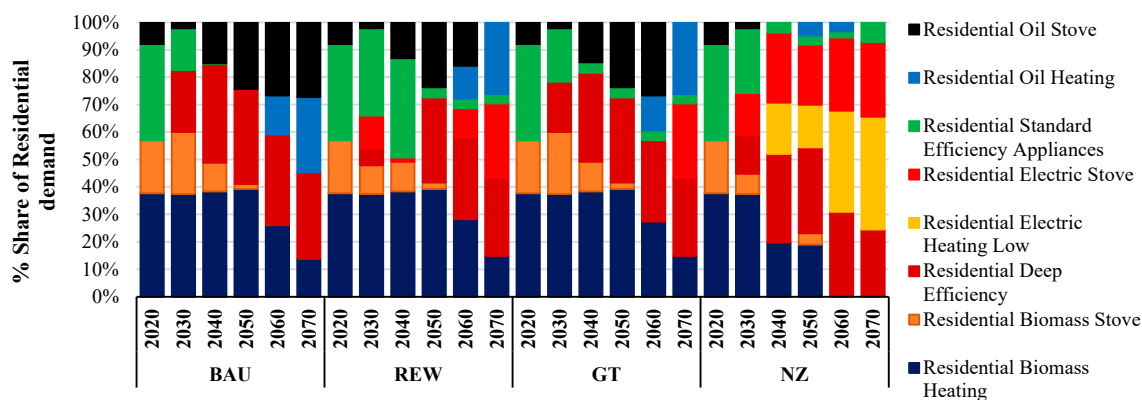


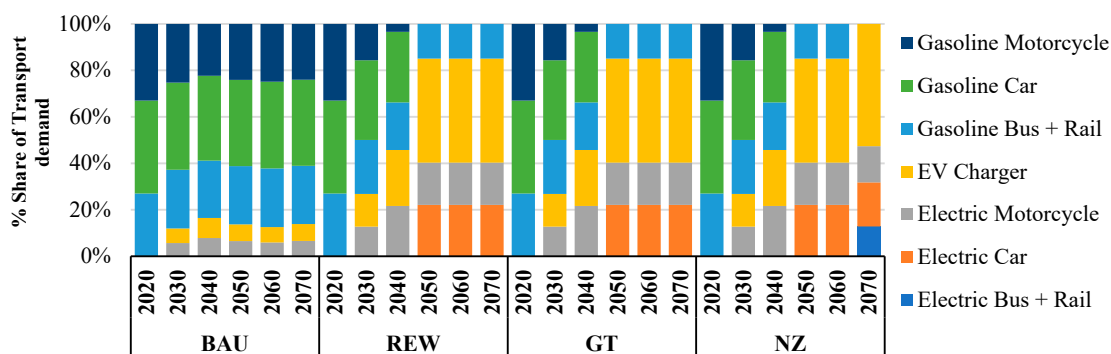
Figure 9. Power generation from 2020 to 2070.

Figure 10 displays the sectoral demand distribution projection across all sectors from 2020 to 2070 under different scenarios. In the BAU scenario, household cooking demand is dominated by traditional stoves and biomass. Conversely, alternative scenarios indicate a shift towards clean cooking and heating technologies. By 2070, the NZ scenario projects that residential cooking will rely primarily on electricity, with electric stoves, heating, and efficiency appliances being the most prevalent. The transport sector transitions from gasoline-based vehicles in the BAU scenario to electricity-based road transportation technologies, with a notable increase in EV chargers' deployment from 2060 onwards. In the industry sector, the REW and GT scenarios show continued demand for oil heating between 2050 and 2060, and biomass heating is projected to represent 25% of industrial demand by 2070. The NZ scenario, however, illustrates a fully decarbonized industry sector by 2070. Moreover, in the commercial sector (Figure 10d), there is practically no difference in energy demand when comparing REW and GT scenarios, with oil for commercial heating accounting for 33.3% of total demand in 2070. The projections underscore the critical role of policy in shaping future energy demand and technology adoption. Policies promoting the transition to clean cooking and heating technologies can significantly reduce reliance on traditional stoves and biomass, aligning with the trends seen in alternative scenarios. The widespread deployment of electric vehicles (EVs) and EV chargers indicates the need for supportive infrastructure and incentives to facilitate this transition. Strategic policy interventions will be crucial in driving the shift towards a sustainable and decarbonized energy future across all sectors.

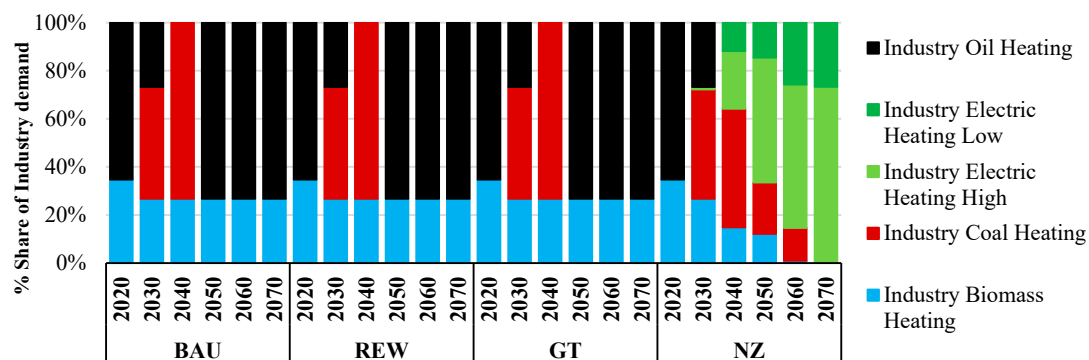
Figures 11 and 12 display the electricity generation and installed capacity for RE and nuclear share in 2050 and 2070 for all scenarios. While only 11% of the electricity is generated from RE technologies under the BAU scenario by 2070, the alternative scenarios show a higher RE integration rate. The share of electricity generated from RE represents 20%, 48%, and 53.5% of the total electricity supply mix in 2070, under GT, REW, and NZ scenarios, respectively. In addition, there is no nuclear power installed in the BAU scenario since there is no currently installed or undergoing nuclear power plant installation in Ghana. Electricity generation from nuclear power represents 5.3%, 33.3%, and 46.5% of the supply mix under REW, GT, and NZ scenarios, respectively, in 2070. These results underscore the crucial role of nuclear power plants in the future energy matrix of the country in the absence of conventional energy generation technologies. In addition to RE technologies, nuclear is a non-GHG emission technology and would ensure a stable power system while emissions are reduced or eliminated, as indicated under the NZ scenario.



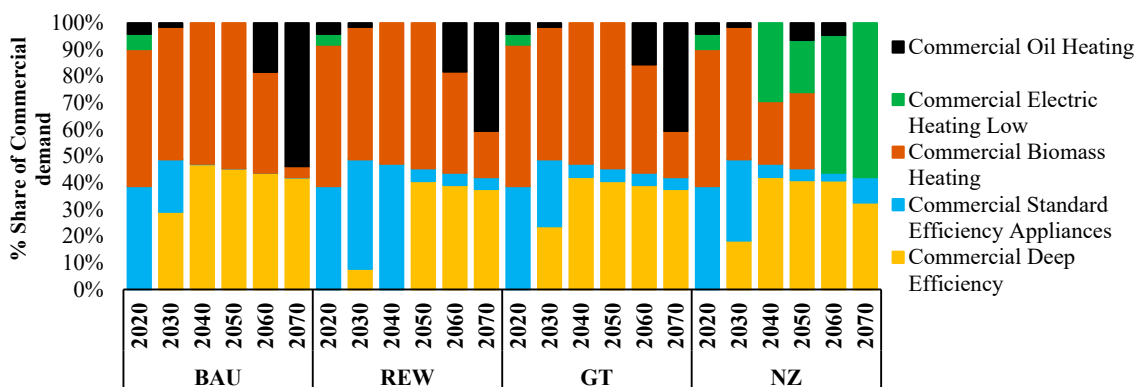
(a) Residential sector



(b) Transport sector



(c) Industrial sector



(d) Commercial sector

Figure 10. Share of sectoral demand for all scenarios: (a) residential energy demand, (b) transport demand, (c) industrial energy demand, (d) commercial energy demand.

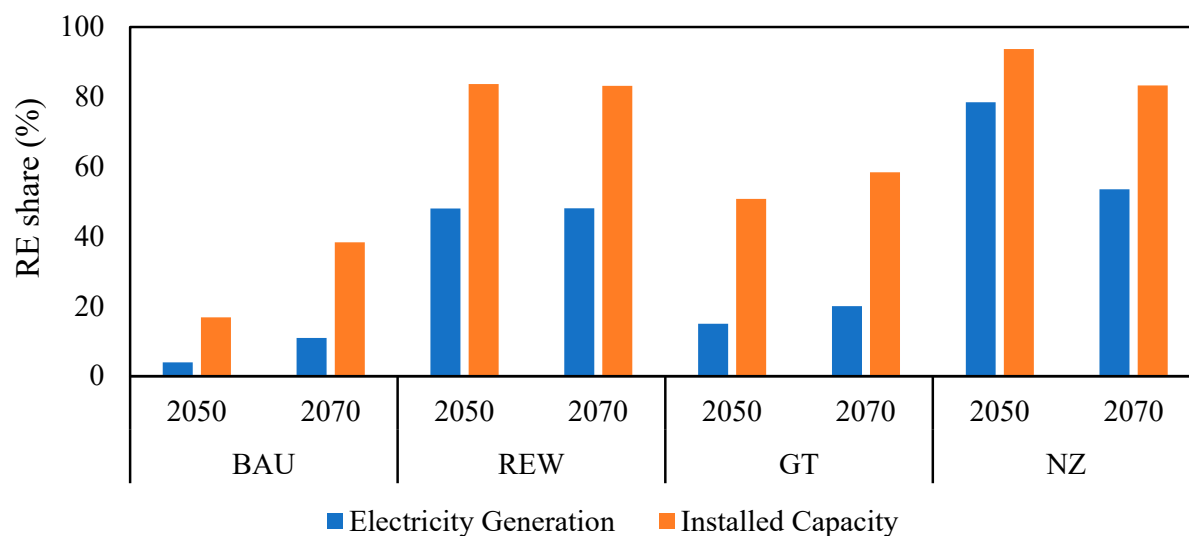


Figure 11. Electricity generation and installed capacity for the RE share in 2050 and 2070.

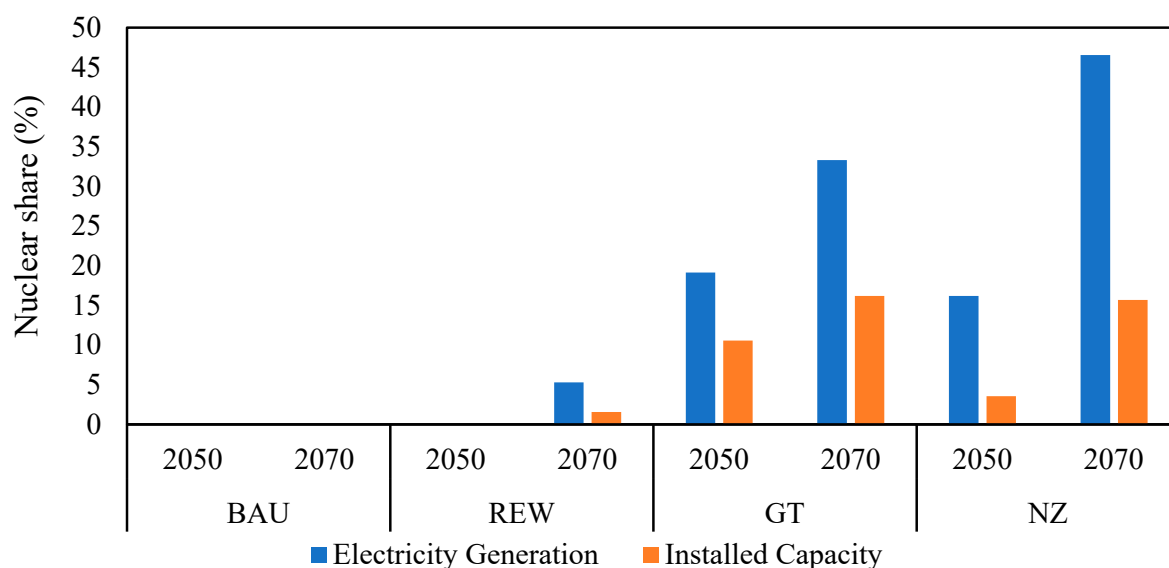


Figure 12. Electricity generation and installed capacity for nuclear share in 2050 and 2070.

3.2. Emission Analysis

Under the BAU scenario, the current trends and pace in the energy sector would continue, indicating a substantial growth of annual emissions. Emissions rose from around 28 MtCO₂ in 2022 to 153.42 MtCO₂ in 2070. The energy demand technologies are still fossil fuel-based. The majority of the emissions come from the transport sector, emitting around 58.2 MtCO₂ in 2070, followed by the power sector (34.1 MtCO₂), industry sector (29.5 MtCO₂), residential sector (26.5 MtCO₂), and commercial sector (5.4 MtCO₂). Figures 13 and 14 illustrate the sector-wise emissions and total CO₂ emissions for all scenarios from 2022 to 2070. The GT and REW scenarios show similar emissions patterns, with a total emission of 105 MtCO₂ in 2070. The transport sector has seen a considerable decrease in GHG emissions compared to the BAU scenario due to the deployment of electric vehicles and EV chargers. However, it is worth mentioning that, in the transport sector, only gasoline-based transport technologies have been modelled in this study. Due to the lack of adequate data, diesel-based road technologies are not modelled, which may lead to low emissions in the transport sector. In addition, the NZ scenario has an emission peak of 37 MtCO₂ in 2032 and decreases linearly to 0 in 2070 as illustrated by Figure 13. The results

demonstrate that the diversification of the energy mix and the integration of clean and sustainable energy technologies could reduce the total emissions and help Ghana achieve its SDG7.

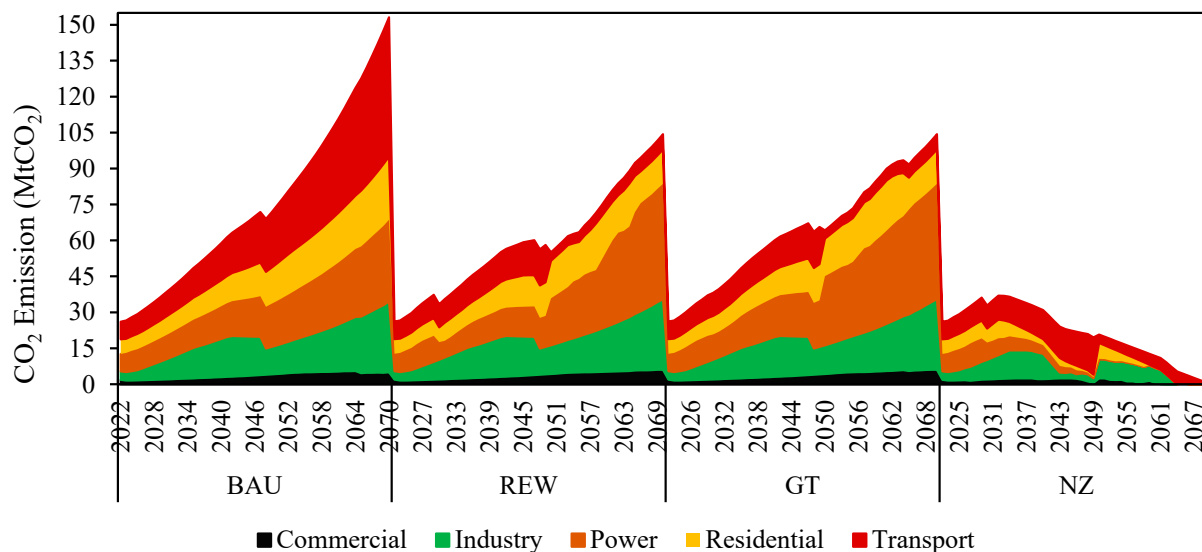


Figure 13. CO₂ emissions by sector for all scenarios from 2022 to 2070.

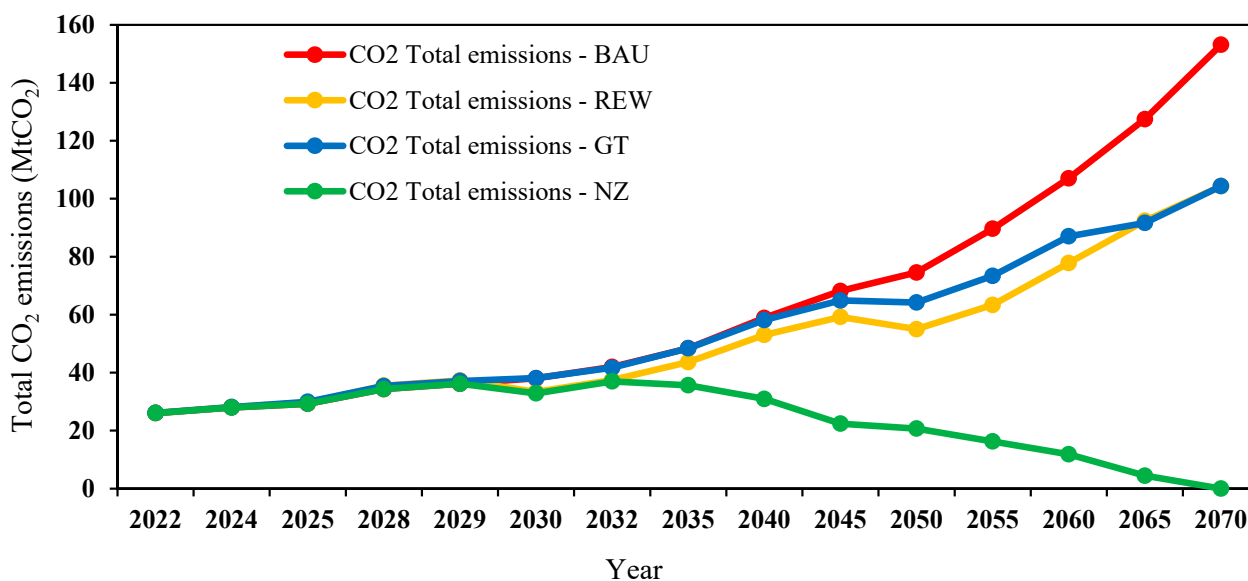


Figure 14. Total annual CO₂ emissions for all scenarios from 2022 to 2070.

3.3. Cost Analysis

Figure 15 presents a comparison of annual fixed operating cost, annual variable operating cost, and capital investment for all scenarios from 2020 to 2070. It can be seen that the annual variable operating cost is the prevalent cost under the BAU scenario. It represents more than 80% of the total annual cost in 2070. However, the alternative scenarios showed a shift from variable cost to an increase in capital investment. The total discounted cost of all scenarios is presented in Figure 16. The discounted cost is the summation of the capital cost, the fixed cost, and the variable cost of the energy system. The total annual cost is estimated to be USD 123.67 billion, USD 108.77 billion, USD 109.86 billion, and USD 112.12 billion under the BAU, GT, REW, and NZ scenarios, respectively. Compared to the BAU scenario, the GT and REW scenarios show cost reductions of approximately USD 14.47 billion and USD 13.38 billion, respectively, indicating that reliance on conventional

fossil fuel technologies would be more expensive than transitioning to renewable energy (RE) technologies by 2070. In addition, comparing the GT scenario, with up to 20% RE and 33.3% nuclear by 2070, and the REW scenario, with at least 48% RE and 5.3% nuclear, the cost reductions suggest that investing in nuclear power would be more beneficial than intermittent RE resources. In addition, the NZ scenario even shows a cost reduction of about USD 11.12 billion, showing that the transition to net zero would become cheaper in the Ghanaian context due to the low cost of transition fuel in the future. This implies that, with adequate policy and energy planning that include both RE technologies and nuclear power, Ghana would achieve net zero by 2070 with less investment.

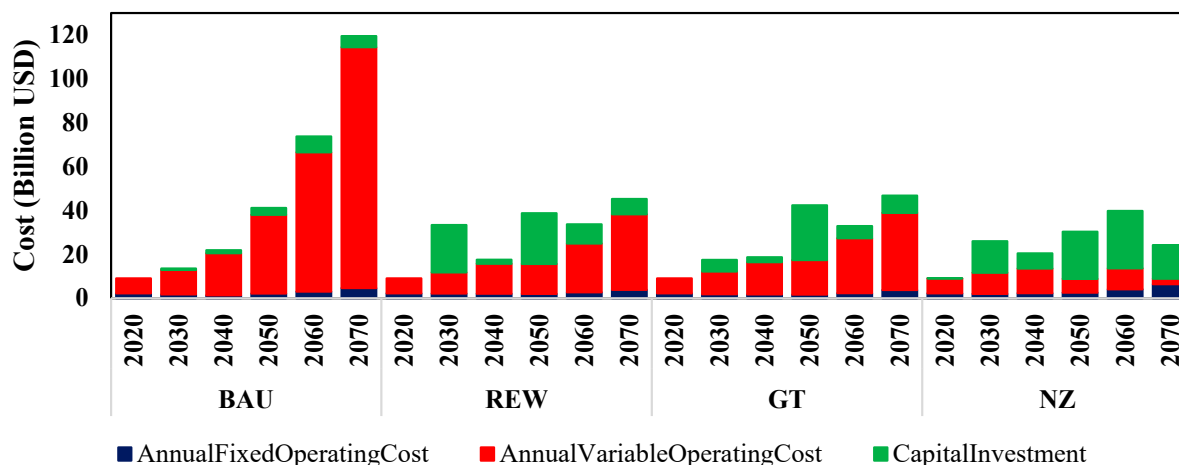


Figure 15. Fixed, variable operating cost, and capital investment.

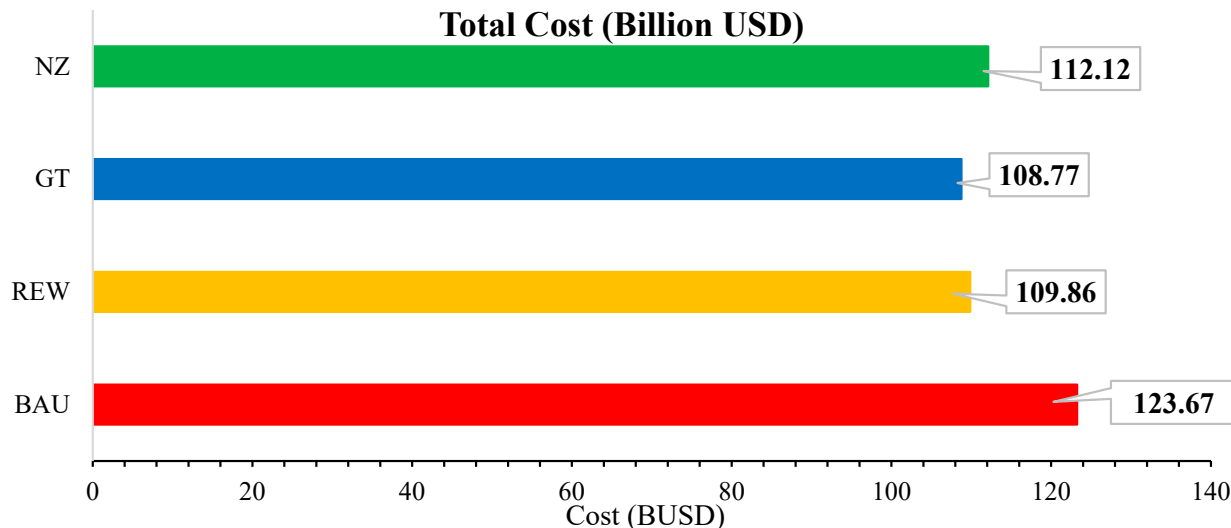


Figure 16. Total cost comparison.

The outcomes indicate that strategic policy interventions and energy planning are crucial for reducing costs and achieving sustainability goals. Policies that promote increased capital investment in renewable energy and nuclear power can significantly reduce long-term operating costs and total system costs. The substantial cost reductions in the GT and REW scenarios underscore the financial benefits of transitioning from fossil fuels to cleaner energy sources. Furthermore, the greater cost savings in the NZ scenario highlight the economic advantages of achieving net-zero emissions, suggesting that comprehensive policies supporting both renewable energy and nuclear power will enable Ghana to achieve its net-zero targets by 2070 with lower investment costs. Implementing these policies will

not only help reduce costs but also promote energy security, environmental sustainability, and economic stability.

3.4. Sensitivity Analysis Results

Table 5 presents the total discounted cost for all scenarios as the discount rate increases. Under the BAU scenario, the total cost decreases from USD 368.4 billion to USD 66.58 billion when the discount rate increases from 5% to 15%. The results indicate that, across all scenarios, an increase in the discount rate leads to a decrease in the total cost. This trend is explained by the OSeMOSYS discounted cost formula, as shown in Equations (1) and (2). According to Garcia-Gusano et al. [65], a higher discount rate would result in a smaller effect on future market extra costs. The findings are also consistent with those of Yassin et al. [66], where a reduction in the discount rate from 22% to 2% results in more favorable market conditions. Figures 17 and 18 illustrate the power generation for discount rates of 5%, 10%, and 15% under the BAU and NZ scenarios, respectively. In the BAU scenario, there is no change in power generation from technologies, such as natural gas plants, solar PV, and large hydropower plants. However, as the discount rate increases, biomass power plants are displaced. In the NZ scenario, onshore wind is partially replaced by utility-scale solar PV and solar PV with 2-h storage. This suggests that higher discount rates do not favor the expansion of biomass power plants but rather support the growth of solar PV technologies. Consequently, the government may need to mobilize investment in solar PV technology.

Table 5. Total annual cost for different discount rates.

	Total Annual Discounted Cost (Billion USD/Year)				
	5%	8%	10%	12%	15%
BAU	368.42	177.74	123.24	92.30	66.58
REW	259.28	147.33	109.86	86.42	65.013
GT	257.75	146.46	108.77	85.26	64.02
NZ	256.48	149.91	112.12	87.99	65.75

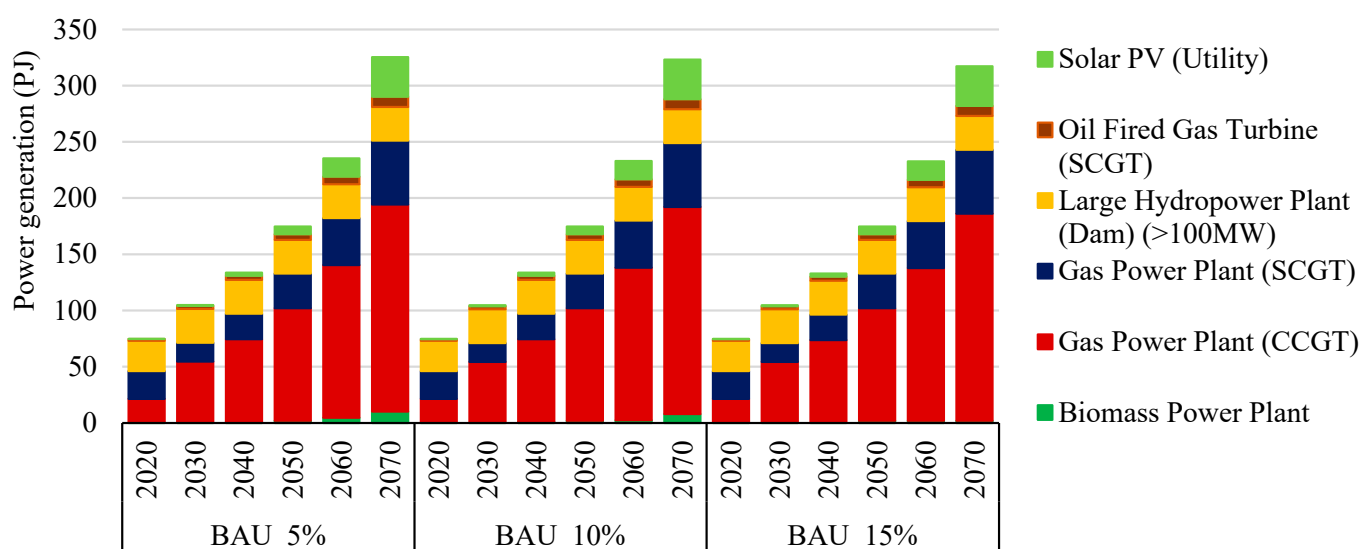


Figure 17. Power generation for 5%, 10%, and 15% under the BAU scenario.

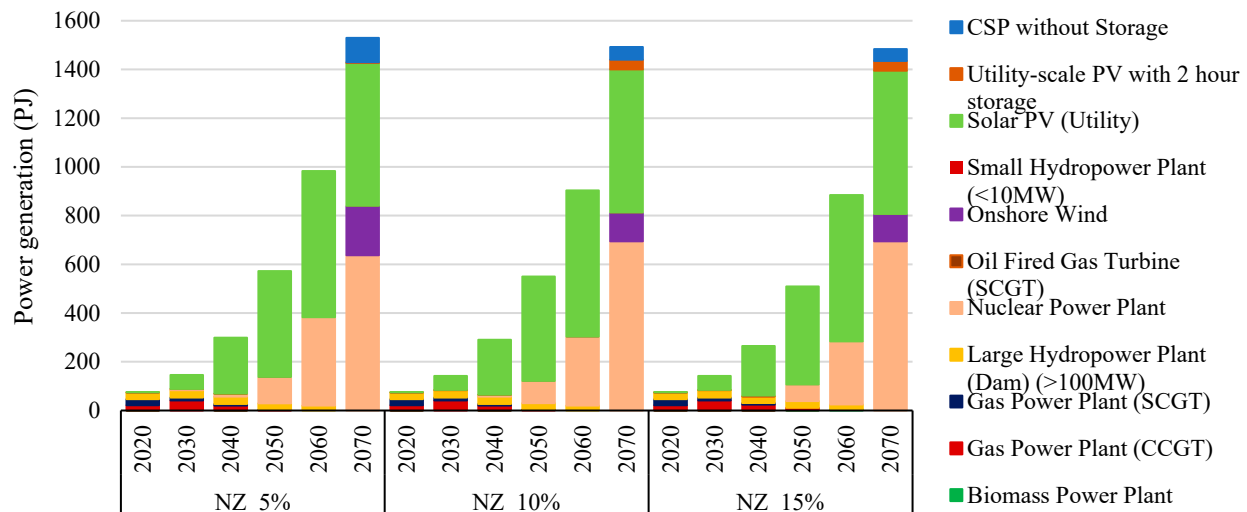


Figure 18. Power generation for 5%, 10%, and 15% under the NZ scenario.

3.5. Model Validation

To verify that the OSeMOSYS-Ghana model is performing as expected, the energy balance of the output is checked based on the RES developed for Ghana in Figure 6. For instance, the total electricity generation is estimated to be 80.77 and 84.6 PJ in 2021 and 2022, respectively, under the BAU scenario. The current values are 79.38 and 83.38 PJ in 2021 and 2022 [16,18], respectively. The error estimation is presented in Table 6. The errors are calculated by taking the difference between the modelled value and the current value, divided by the current value, and multiplied by 100. The trend is also verified for the installed capacity and total annual GHG emissions, with a total emission of around 28 MtCO₂ equivalent estimated for 2022. The BAU scenario results are also compared to the No Policy Intervention (NPI) scenario of Ghana, as presented in their energy transition and investment plan report, to validate the model [3].

Table 6. Model validation.

	Generation (PJ)	
	2021	2022
Current	79.38	83.38
Modelled	80.66	84.45
Error (%)	1.612	1.283

4. Discussion

4.1. General Discussions and Policy Implications

The results indicate a substantial increase in Ghana's power generation capacity by 2070 under the alternative scenarios (GT, REW, and NZ) compared to the BAU scenario. This substantial growth is primarily due to the integration of renewable energy (RE) technologies, such as solar PV, onshore wind, and concentrated solar power (CSP), to power all sectors, while meeting end-use demand with electricity. The transition towards RE technologies observed in the alternative scenarios highlights the importance of diversifying the energy supply mix to ensure reliability and sustainability. This trend is consistent with findings from other regions, such as Germany and California, where large-scale integration of RE has led to significant increases in installed capacities to maintain energy security and meet demand [67]. In other developing regions, such as Egypt and Ethiopia [21,66], RE

resources have been found as potential energy sources to meet future energy demand and reduce emissions.

The projected shift in energy demand across sectors underscores the transformative impact of transitioning to clean energy technologies in Ghana's energy system. By 2070, the NZ scenario anticipates a complete reliance on electricity for residential cooking, with electric stoves and efficient appliances becoming predominant. This transition is also observed in the transportation sector, where a move from gasoline-based vehicles to electric vehicles (EVs) is expected, significantly reducing emissions. Similar trends have been observed in Scandinavian countries [68] and Colombia [8,20], where government policies and incentives have shown potential to accelerate the adoption of EVs and electric heating technologies. These findings suggest that Ghana can achieve substantial health and environmental benefits through targeted policies promoting clean cooking and transportation technologies. The role of renewable energy and nuclear power in the future energy mix is highlighted by the significant contributions of these technologies in the alternative scenarios. By 2070, renewable energy accounts for up to 58.3% of the electricity supply in the NZ scenario, while nuclear power provides a stable, low-emission source of energy. The significant contribution from the nuclear power plant to the Ghana energy mix by mid-century demonstrated by the alternative scenarios, underscores its importance in achieving a diversified and resilient energy system. This mirrors the experiences of countries, like France and South Korea, where nuclear power has played a crucial role in reducing greenhouse gas emissions and ensuring energy security. These results suggest that strategic investments in nuclear and RE technologies are essential for the country to meet its emissions reduction targets.

The Ghana energy transition framework [3,64], suggests an RE integration of 10% by 2030 and 20% by 2070, which is explored in the GT scenario. However, it is clear that, to achieve net zero by 2070, a higher rate of RE is advisable. The NZ scenario, for instance, showed about 53% of RE by 2070 to complement nuclear power in the future energy mix. Similar to the present study, refs. [28,69] have investigated decarbonization pathways for Dominica and Indonesia, respectively. However, these studies did not include the contribution of nuclear power plants to the future energy system. The economic analysis reveals significant cost savings in the alternative scenarios compared to the BAU scenario. By 2070, the NZ scenario is projected to save approximately USD 11.12 billion compared to the BAU scenario, emphasizing the financial viability of transitioning to a net-zero energy system. Studies in the United Kingdom support this finding, and Denmark, which is transitioning to renewable and low-carbon technologies, has reduced overall energy system costs [70], and in Benin, where renewable energy has the potential to boost sustainable energy transition. The findings are also confirmed by Albert et al. [13], who used OSeMOSYS to simulate technology development scenarios and evaluate RE integration approaches in Ghana. A similar approach to this study is used in Ethiopia [21], Columbia [8], and the Democratic Republic of Congo (DRC) [10] to assess different levels of renewable energy integration into the supply mix.

Additionally, the substantial reduction in emissions, particularly in the NZ scenario, which achieves net zero by 2070, highlights the environmental benefits of adopting clean energy technologies. These results underscore the importance of comprehensive energy planning and policy support to drive the transition towards a sustainable and low-carbon future in Ghana. According to the International Energy Agency (IEA) [71], the total investment flows toward clean energy, including RE and nuclear, reached USD 1.9 trillion in 2023. This indicates how financial resources are being mobilized towards the deployment of RE and nuclear technologies, and therefore, developing countries, like Ghana, should

prioritize investment in such systems for an affordable, reliable, and sustainable future energy system.

This study provides critical insights into developing long-term emissions strategies (LTES) and enhances the Data-to-Deal (D2D) process for Ghana by offering a comprehensive workflow to call for crucial investment and facilitate the reduction of long-term emissions through analysis of various scenarios [72]. It is indispensable for overcoming barriers to climate finance, crafting an adequate policy framework, and establishing a transparent financing mechanism for the country [3]. By employing a modelling approach that integrates local data, policy objectives, and socioeconomic parameters, the study presents a robust framework of options aimed at strengthening Ghana's core capabilities at the national level. It is crucial to underscore that this energy transition demands massive transformations and the adoption of cutting-edge technologies across multiple sectors. The analysis reveals that investment costs are lower in the alternative scenarios compared to the BAU scenario, highlighting the greater long-term benefits and cost savings. Notably, the fully decarbonized scenario (NZ) presents a greater reduction in terms of GHG emissions and energy efficiency improvement in the long term.

The open-source energy modelling approach adopted in this study is crucial for fostering transparency, collaboration, and innovation in the energy sector. It allows for the verification and reproduction of results, ensuring robust and reliable outcomes that are essential for informed policy-making. By being accessible and free, open-source energy models democratize access to advanced modelling tools, enabling a wider range of participants, including developing countries, to engage in energy analysis. This adaptability and community-driven development make the open-source modelling approach versatile and continually improving, supporting a more inclusive and sustainable global energy transition. This study's modelling approach is transparent for stakeholders' discussion and easy for collaborative work across institutions.

Hence, the study results demonstrate that transitioning to renewable and nuclear energy technologies can significantly increase Ghana's power generation capacity, reduce emissions, and provide substantial economic savings. The findings align with similar studies in other regions, reinforcing the importance of diversifying the energy supply mix and implementing supportive policies to achieve sustainable development goals. Strategic investments in clean energy technologies and infrastructure are crucial for Ghana to meet its future energy demands while ensuring environmental sustainability and economic viability. In addition, the significance of the findings relies on the importance of strategic planning and investment in renewable energy infrastructure to realize these future energy scenarios. The findings are useful to decision-makers and stakeholders in the energy sector to develop appropriate policies that would promote the development of clean energy technologies for Ghana's socioeconomic development. Future policy actions should include:

Immediate and short term: prioritizing investments in RE infrastructure and technologies to capitalize on decreasing costs and technological advancements. Implementing supportive policies for energy efficiency and demand-side management would reduce overall consumption and operational costs.

Medium to long Term: developing nuclear power as a stable baseload energy source, ensuring a reliable energy supply while integrating more renewables; and developing diverse financing mechanisms, including green bonds, international grants, and public-private partnerships, to mobilize necessary investments.

4.2. Limitations and Further Work

The modeling results are generally consistent with the Ghana National Energy Transition Framework, previous energy transition projects, and existing research on OSeMOSYS

and energy transitions in developing countries. However, numerical outcomes may vary due to differences in models, calculation methods, and datasets, highlighting areas for further refinement. A primary challenge arises from the limited availability of detailed data on technology costs, energy consumption, and demand patterns within different end-use sectors. The absence of data poses a significant challenge to performing accurate analyses. Establishing data collection institutions and publicly accessible data websites is essential for researchers. In addition, the assumption made in this study may not always reflect reality. As this study is not doing any forecasting, it is suggested that future work combine population growth and other demographic factors, including the rate of economic growth and structural changes in the economy, to project long-term demand. Further research could investigate the possibilities of integrating green hydrogen technologies in the transport sector of the country. Furthermore, the flexibility assessment of the developed scenarios to integrate a high share of RE into the national grid could constitute a research direction for future studies. Additionally, adopting a Climate–Land–Energy–Water (CLEW) system approach would be valuable for analyzing relationships between environment, land, water, and energy (nexus) through systems, like waste-to-energy (W2E), agricultural irrigation, and land use, thereby aiding in the development of robust net-zero policies. Generally, it is well known that EV adoption is also driven by consumer behavior as well as infrastructure readiness. However, this study does not incorporate this or its impacts on GHG emissions. It is therefore recommended that future studies analyze consumer behavior and infrastructure readiness in the Ghanaian context for EV adoption. A future study could collect up-to-date road transportation data, including diesel consumption for buses and trucks, to conduct further analysis to enhance the transport sector modelling analysis conducted in this study.

5. Conclusions

This study investigates strategies for long-term energy transition analysis in the Ghanaian context. Four energy scenarios have been developed, namely the Business as Usual (BAU), the 48% RE integration (REW) scenario, the Government Target (GT) scenario, and the Net-Zero (NZ) scenario. The study identifies the optimal supply mix alternatives for Ghana's future power system to ensure adequate, reliable, and sustainable energy generation while demonstrating the potential role of nuclear power as a clean and reliable energy source. The OSeMOSYS-Gh model was employed to explore the developed scenarios and analyze results based on the total power generation, the installed capacity, total annual cost, and associated GHG emissions. The main findings are as follows:

- Total energy generation and the installed capacity are expected to increase in all scenarios, with REW, GT, and NZ scenarios shifting towards more renewable energy and nuclear resources. Under the BAU scenario, natural gas remains the dominant energy source, comprising about 60% of power generation by 2070. The alternative scenarios, however, depict a more diversified energy mix, incorporating solar photovoltaic (PV), onshore wind, concentrated solar power (CSP), small hydropower, and nuclear power. By 2070, the share of electricity production from natural gas would be reduced to 31% in both REW and GT scenarios, with nuclear power becoming a significant contributor beyond 2040 in both scenarios. In the NZ scenario, the total installed capacity and annual power generation are estimated to be 168 GW and 1500 PJ, respectively, in 2070. The share of electricity generation is dominated by RE technologies at 53.6%, and the remaining (46.4) is constituted by nuclear power production.
- In contrast to the BAU scenario, the alternative scenarios had the advantage of electrifying the end-use sectors. If the REW and GT scenarios present similar demand patterns in the different sectors, NZ shows a fully decarbonized system by 2070. Total

GHG emissions rose from around 28 MtCO₂ in 2022 to 153.2 MtCO₂ in 2070 under the BAU scenario. Moreover, each of the GT and REW scenarios presents an emissions reduction of 48.2 MtCO₂ compared to the BAU scenario.

- In terms of the total annual cost, the REW, GT, and NZ scenarios are more favorable, with lower total discounted costs compared to the BAU scenario. The NZ scenario shows a cost reduction of approximately USD 11.55 billion, while the GT and REW scenarios offer a greater reduction of about USD 14.90 billion and USD 13.81 billion, respectively, compared to the BAU scenario. These cost savings underscore the financial advantages of transitioning to renewable and nuclear energy technologies.

This study employs an open-source modeling approach, offering a replicable framework for energy transition analysis in Ghana and other developing countries. The use of an open-source approach provides transparency and adaptability, enabling policymakers and researchers to explore optimal pathways for energy planning with limited resources. The inclusion of nuclear energy as a clean, reliable energy source is critical, emphasizing its potential role in achieving a diversified and resilient energy mix. To realize these outcomes, Ghana should adopt robust policies and regulatory frameworks to support the integration of renewable and nuclear energy technologies. These policies should focus on increasing electrification, enhancing energy efficiency, and attracting investments in sustainable energy infrastructure. Maintaining natural gas as a transition fuel can provide stability during this evolution. By embracing these strategies, Ghana can build a resilient energy sector that addresses population needs, fosters economic growth, and aligns with global climate goals. Moreover, the methodology and scenarios developed in this study offer valuable insights for other developing countries with similar energy contexts, reinforcing the global applicability of this research. The sensitivity analysis conducted suggested that the government may need to mobilize investment in solar PV technology. The sensitivity analysis conducted as part of this research underscores the importance of mobilizing investments in solar PV technologies, given their cost-effectiveness and potential for large-scale deployment in Ghana's energy landscape. Beyond Ghana, this modelling approach is highly applicable to other developing countries, especially those facing data limitations and similar development challenges. It provides a practical framework for researchers to explore evidence-based national and regional strategies that can accelerate sustainable energy transitions and support long-term development planning in resource-constrained contexts.

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U4RIA Compliance Statement: This work follows the U4RIA guidelines, which provide a set of high-level goals related to conducting energy system analyses in countries. This paper was carried out involving stakeholders in the development of models, assumptions, scenarios, and results (Ubuntu/Community). The authors ensure that all data, source code, and results can be easily found, accessed, downloaded, and viewed (retrievability), licensed for reuse (reusability), and that the modelling process can be repeated in an automatic way (repeatability). The authors provide complete metadata for reconstructing the modelling process (reconstructability), ensuring the transfer of data, assumptions, and results to other projects, analyses, and models (interoperability), and facilitating peer review through transparency (auditability).

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Abbreviations

The following abbreviations are used in this manuscript:

FD	Final Energy Demand
FECT	Fuel Energy Consumption per Technology
GDP	Gross Domestic Product
GHG	Greenhouse Gas Emissions
GRIDCo	Ghana Grid Company
GT	Government Target
GW	Gigawatt
GWP	Global Warming Potential
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
LPG	Liquefied Petroleum Gas
MW	Megawatt
NDCs	Nationally Determined Contributions
NG	Natural gas
NPI	No Policy Intervention
NZ	Net-zero
OSeMOSYS	Open-Source Energy Modelling SYStem
PJ	Petajoules
PV	Photovoltaic
RE	Renewable Energy
RES	Reference Energy System
REW	RE Integration
SAND	Simple And Nearly Done
SCGT	Single Cycle Gas Turbine
SDG	Sustainable Development Goals

TRC	Technology Residual Capacity
USD	United States Dollar
W2E	Waste-To-Energy
y	Year in Time Horizon
t	Type of Power Plant Technology
r	Region or Country

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